TIRARIT

TREE IN RAYO AND RAYO IN TREE

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Abstract

TIRARIT is a mathematical construction based on the composition and repeated application of two mathematical functions: the TREE function and a generalized Rayo function, denoted by R.

The central idea is to combine these functions into a single function and repeatedly apply that function, starting from TREE(3).

This paper introduces the TREE function, explains Rayo's Number, defines the generalized function R, presents the mathematical definition of TIRARIT, and proves that its first value exceeds the Rayo benchmark under the stated symbol-counting convention.

The main result is:

$$\mathbf{TIRARIT(1) > R(G)}$$

where $G = 10^{100}$.

1. Introduction

Mathematics contains many extremely large finite numbers. Some of them are so large that writing their complete decimal expansions is practically impossible.

Examples include Graham's Number, TREE(3), and Rayo's Number. These numbers are defined using different mathematical ideas.

TIRARIT combines two important concepts:

- The TREE function, originating from combinatorics and graph theory.
- The generalized Rayo function, based on the limits of mathematical definability.

The purpose of TIRARIT is to construct a new function by repeatedly composing these two functions.

2. The TREE Function

The TREE function is a mathematical function related to Kruskal's Tree Theorem. It is defined using finite rooted trees whose vertices are labeled with a finite number of possible labels.

For TREE(k), consider sequences of finite rooted trees satisfying the following conditions:

1. Each tree uses one of k possible labels.

2. The i-th tree contains at most i vertices.
3. No earlier tree can be embedded into a later tree under the standard label-preserving tree embedding relation.

TREE(k) is the maximum possible length of such a sequence. Kruskal's Tree Theorem guarantees that this maximum is finite.

2.1. Initial Values

The first two values are:

$$TREE(1) = 1$$

$$TREE(2) = 3$$

However, TREE(3) is extraordinarily large.

2.2. Why TREE(3) Is Important

TREE(3) is a finite integer that is larger than Graham's Number. Graham's Number is itself much larger than a googol:

$$G = 10^{100}$$

Therefore:

$$TREE(3) > G$$

TREE(3) is a particularly famous example of an extremely large finite mathematical number.

3. Rayo's Number

Rayo's Number was introduced during a large-number competition in 2007. Its underlying idea concerns mathematical definability.

A natural number is definable if a mathematical expression in a formal language uniquely identifies that number. However, the number of symbols available for writing an expression can be restricted.

Rayo's construction considers all natural numbers that can be uniquely defined using fewer than a specified number of symbols. The number is then chosen to be greater than every natural number definable within that limit.

The famous Rayo benchmark uses a symbol limit based on a googol:

$$G = 10^{100}$$

Rayo's Number is finite, despite its extraordinary size. Its definition relies on formal mathematical language rather than ordinary decimal notation.

4. The Generalized Rayo Function R

TIRARIT uses a generalized function based on the idea behind Rayo's Number. We define:

$$R(n) = 1 + \max(D_n)$$

where D_n is the set of all natural numbers uniquely definable using fewer than n symbols in a fixed formal language.

In simpler terms: R(n) is the smallest natural number greater than every natural number uniquely definable using fewer than n symbols.

4.1. Difference Between R and Rayo's Number

Rayo's Number is a particular number. R is a function that accepts a symbol limit as its input.

For example, R(100) is greater than every natural number uniquely definable using fewer than 100 symbols.

Under the strict convention of fewer than G symbols, the corresponding Rayo benchmark is R(G). If a definition instead uses at most G symbols, the corresponding input under our convention is G + 1.

4.2. Monotonicity of R

The function R is nondecreasing. For any natural numbers a and b:

$$\text{If } a \leq b, \text{ then } D_a \subseteq D_b$$

This is because every expression containing fewer than a symbols also contains fewer than b symbols. Therefore:

$$R(a) \leq R(b)$$

This property is essential for the main proof.

5. Definition of TIRARIT

The name TIRARIT stands for: **TREE IN RAYO AND RAYO IN TREE**.

First, define a new function:

$$F(x) = \text{TREE}(R(x))$$

Then define TIRARIT as:

$$\text{TIRARIT}(n) = (\text{TREE} \blacksquare R)^n(\text{TREE}(3))$$

Here, the symbol ■ represents function composition. The exponent n represents repeated function application, not ordinary exponentiation. In other words, the function F is applied n times, starting from TREE(3).

5.1. Examples

For n = 1:

$$\text{TIRARIT}(1) = \text{TREE}(R(\text{TREE}(3)))$$

For n = 2:

$$\text{TIRARIT}(2) = \text{TREE}(R(\text{TREE}(R(\text{TREE}(3)))))$$

For n = 3:

$$\text{TIRARIT}(3) = \text{TREE}(R(\text{TREE}(R(\text{TREE}(R(\text{TREE}(3))))))))$$

The same process continues for larger values of n. For example, TIRARIT(160) means applying F 160 times to TREE(3).

6. Proof That TIRARIT(1) Exceeds the Rayo Benchmark

We now prove the following inequality:

$$\text{TIRARIT}(1) > R(G)$$

Recall that $G = 10^{100}$ and TIRARIT(1) = TREE(R(TREE(3))). The proof uses the monotonicity properties of R and TREE.

6.1. Step One: Comparing TREE(3) with G

It is known that TREE(3) > Graham's Number. Graham's Number is greater than a googol. Therefore:

$$TREE(3) > G$$

6.2. Step Two: Applying the Monotonicity of R

Since R is nondecreasing and TREE(3) > G:

$$R(TREE(3)) \geq R(G)$$

Let X = R(G). Then:

$$R(TREE(3)) \geq X$$

6.3. Step Three: A Lower Bound for TREE

We use the following property of the standard TREE function:

$$TREE(n) \geq n + 1 \text{ for every } n \geq 2.$$

To establish this bound, consider a valid sequence of trees using k labels. A new label can be introduced, and a single-vertex tree carrying that new label can be placed at the beginning of the sequence. Because the new label does not occur in the remaining trees, the new first tree cannot embed into any later tree. The previous sequence remains valid. Consequently:

$$TREE(k + 1) \geq TREE(k) + 1$$

Starting with TREE(2) = 3 we obtain TREE(n) ≥ n + 1 for every n ≥ 2. Therefore:

$$TREE(X) > X$$

6.4. Step Four: Applying the Monotonicity of TREE

From Step Two:

$$R(TREE(3)) \geq X$$

Since TREE is increasing:

$$TREE(R(TREE(3))) \geq TREE(X)$$

From Step Three: TREE(X) > X. Combining these inequalities gives:

$$TREE(R(TREE(3))) > X$$

Substituting X = R(G):

$$TREE(R(TREE(3))) > R(G)$$

Finally, by the definition of TIRARIT(1):

$$\mathbf{TIRARIT(1) > R(G)}$$

This completes the proof.

7. Mathematical Interpretation

TIRARIT combines two different mathematical mechanisms. The generalized Rayo function R uses the expressive limits of a formal language. The TREE function uses the combinatorial complexity of finite labeled trees.

Their composition creates a function that can be repeatedly applied to its own previous output. The result is a hierarchy of increasingly large finite integers.

The important feature is that the first value already exceeds the Rayo benchmark under the stated convention.

8. Notation and Definitions

The following notation is used throughout this paper:

G — The googol: $G = 10^{100}$

D_n — The set of natural numbers uniquely definable using fewer than n symbols in the fixed formal language.

R(n) — The generalized Rayo function: $R(n) = 1 + \max(D_n)$

TREE(n) — The standard TREE function defined through finite labeled rooted trees and the tree embedding relation.

F(x) — The composition: $F(x) = TREE(R(x))$

TIRARIT(n) — The n-fold iteration of F starting from TREE(3): $TIRARIT(n) = F^n(TREE(3))$

For a fully formal implementation, the language, semantics, unique-definability conditions, and symbol-counting rules must remain fixed throughout the definition.

9. Conclusion

TIRARIT is a mathematical construction based on the repeated composition of the TREE function and the generalized Rayo function.

Its definition is:

$$TIRARIT(n) = (TREE \blacksquare R)^n(TREE(3))$$

The main comparison established in this paper is:

$$\mathbf{TIRARIT(1) > R(G)}$$

This follows from:

- TREE(3) being greater than a googol.
- The nondecreasing property of R.
- The lower bound $TREE(n) \geq n + 1$ for $n \geq 2$.
- The increasing behavior of TREE.

TIRARIT demonstrates how two distinct mathematical ideas can be combined to construct an extremely large hierarchy of finite numbers.

References

1. Kruskal's Tree Theorem.
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3. Wolfram MathWorld, Rayo's Number.
4. Rayo's Big Number Duel (2007).
5. Harvey Friedman's writings on finite trees and large numbers.

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